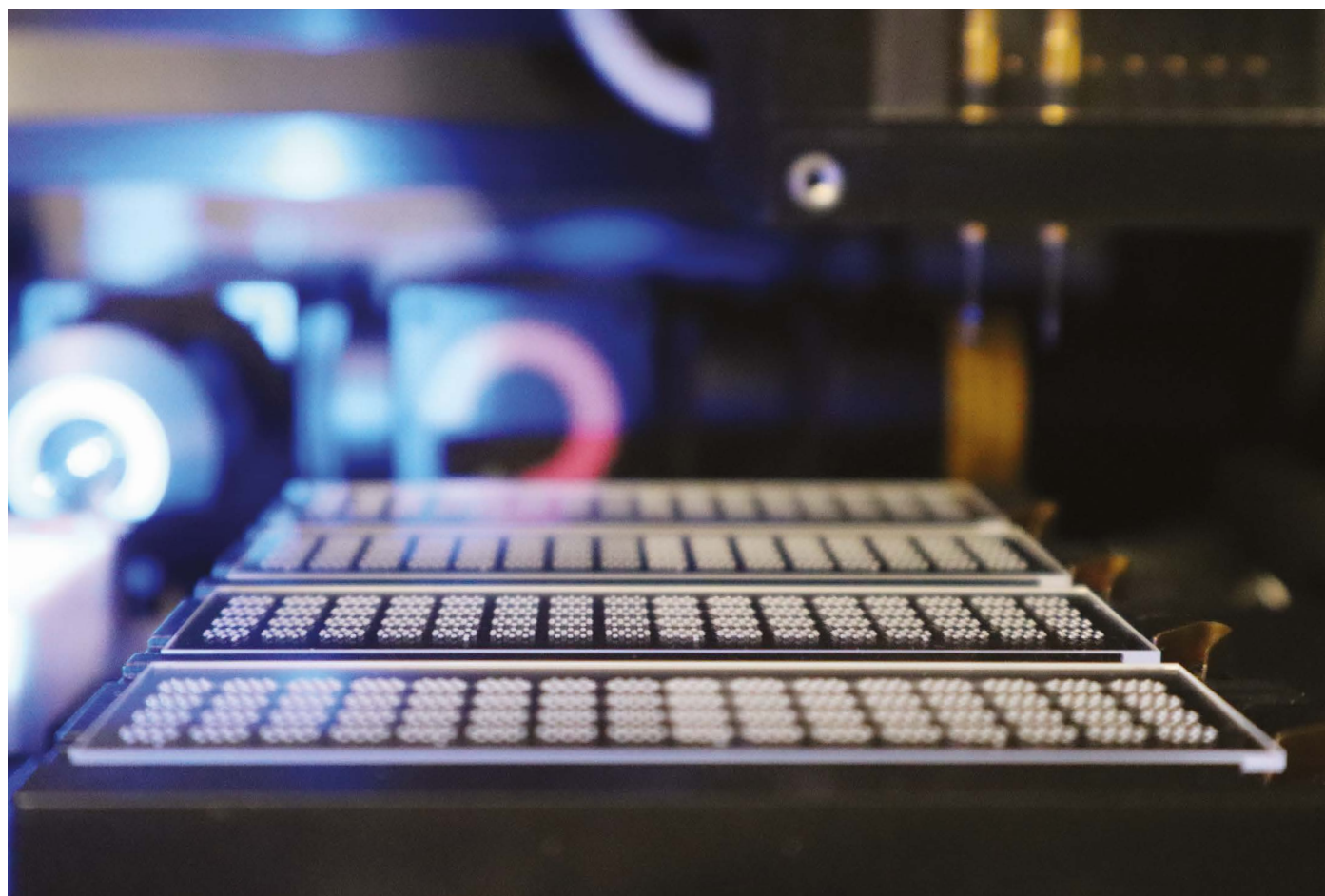




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Single-cell samples arrayed on glass slides for analysis by mass spectrometry.

PROBING THE PROTEOME AT CELLULAR SCALE

It's now feasible to identify thousands of proteins in individual cells, revealing the mechanisms of development and disease. **By Michael Eisenstein**

When a fertilized mammalian egg divides, it yields two seemingly identical daughter cells. Some of their descendants generate the diverse organ systems of the embryo, while others form extra-embryonic tissues that sustain the developmental process.

In 2025, researchers used a cutting-edge technique known as single-cell proteomics (SCP) to show that even in the two first-generation cells, this division of labour is already established¹. Nikolai Slavov at Northeastern University in Boston, Massachusetts, who co-directed the study with fellow cell biologists Magdalena Zernicka-Goetz and Tsui-Fen Chou at the California Institute of Technology in Pasadena, says

that the protein contents of these two cells, which were termed 'alpha' and 'beta', reveal distinct properties and fates. "Beta cells were more likely to give rise to healthy embryos than the alpha cells," he says, whereas alpha cells typically formed extra-embryonic tissues. The team could trace these identities back to fertilization.

Other questions are also yielding to the technology. Researchers now recognize, for instance, that many of the most clinically significant events in the formation of tumours originate from individual cells camouflaged in the chaotic tumour environment. "Single-cell proteomics may help us understand why patients have different outcomes due to tumour evolution, immune responses and cell differentiation – all driven by cellular heterogeneity,"

says Yu-Ju Chen, a mass spectrometrist at Academia Sinica in Taipei.

Just a few years ago, protein biologists largely dismissed the possibility of single-cell proteomics – profiling thousands of proteins in individual cells – as the stuff of science fiction. "I'm almost on the record as saying, 'not in my lifetime', because it seemed to be so far off," says Matthias Mann, a proteomics researcher at the Max Planck Institute of Biochemistry in Martinsried, Germany. But over the past decade, SCP has become not only possible but practical – and scientifically informative. Mann himself is now an avid practitioner, using SCP to study illnesses ranging from liver conditions to Alzheimer's disease. The bar to entry remains high, however, and the need for expensive equipment, deep expertise and

methodological precision to process cell-scale volumes of protein still limits the potentially transformative impact of this technology.

Precious cargo

Biology is already deep in the single-cell era, but most such analyses focus on the transcriptome. Profiling gene expression at the level of the individual cell can provide valuable information about a cell's identity or physiological state. But researchers cannot necessarily predict the protein contents of a cell from its RNA alone – not every RNA gets translated, and numerous factors shape the timing and extent of protein production from a given transcript. Many disease mechanisms can best be understood by surveying the protein contents – for example, the pathology of Alzheimer's and Parkinson's disease is strongly linked to aberrant aggregates of protein. “The proteome reflects the current state of the cell,” Mann says.

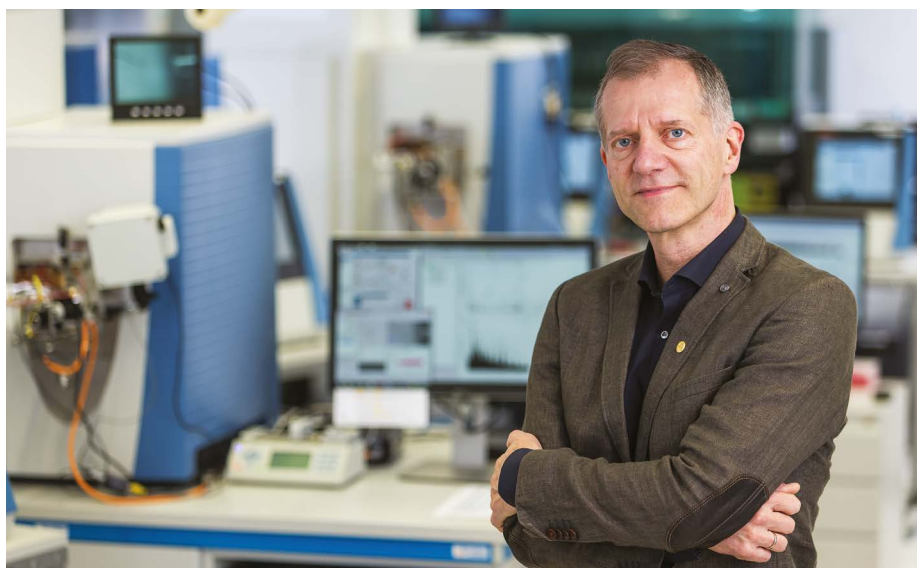
To survey that state, proteomics researchers typically work in bulk, pooling and analysing thousands or millions of cells using mass spectrometry (MS). This technique uses specialized instruments that break samples down into ionized pieces, which are then analysed in terms of their mass and charge. The resulting ‘fingerprint’ enables precise identification of the molecule from which the fragment originated, and researchers can routinely detect 10,000 or more distinct proteins per sample. But this is an averaged perspective that can obscure the underlying biology. “Are they really reflecting what's going on in all cells or is it just a small population of cells where you see dramatic changes?” asks Jesper Olsen, a mass spectrometrists at the University of Copenhagen. “These are things we have not really been able to address before.”

Enter single-cell proteomics

In some ways, the technology should be simpler than its transcriptomic analogue. Protein is more stable than RNA is, and more abundant. “With RNA, we count zero, one, two, three [copies] most of the time; with proteins, we count 100, 500, 10,000,” says Slavov. But transcriptomics researchers can boost the cellular RNA signal by using the PCR amplification technique. No such tool exists for the proteome, and researchers must learn all they can from a few hundred picograms of material.

Getting a robust and informative signal from such tiny quantities requires a highly sensitive mass spectrometer – but sample preparation poses a more formidable challenge. “The raw sensitivity was there ... but the sample prep was optimized for bulk level,” says Ryan Kelly, a biochemist at Brigham Young University in Provo, Utah.

In 2018, a team led by Slavov demonstrated an important first step towards broad proteomic coverage at the single-cell level². The researchers developed a sample preparation



Matthias Mann's Deep Visual Proteomics strategy adds spatial data to single-cell proteomics.

and analysis workflow called SCoPE-MS (single-cell proteomics by mass spectrometry), in which they chemically labelled proteins in individual cells of interest and then mixed these with an excess of unlabelled ‘carrier’ cells. This limited protein loss from the target cells and made protein identification easier, enabling the detection of more than 1,500 proteins across eight individual cells. “We essentially gained about four orders of magnitude in sensitivity,” he says.

Slavov recalls that the publication of his results was met with scepticism, but other groups soon replicated them. And importantly, a second group led by Kelly published

“Single-cell proteomics may help us understand why patients have different outcomes.”

an independent approach to SCP that same year, using a sample preparation device called the nanoPOTS chip (nanodroplet processing in one pot for trace samples) to identify 670 proteins in small numbers of individual cells³. “The whole point of those early papers was, ‘this is real signal from single cells,’” says Kelly. “It’s not a question any more.”

A growing crowd

Since then, the field has advanced at a remarkable pace. Jennifer Van Eyk, a clinical proteomics researcher at Cedars-Sinai Medical Center in Los Angeles, California, says that her team has developed a workflow that can consistently identify 1,500–2,000 proteins per cell. “The biology we’re seeing is just unbelievable,” she says.

And deeper coverage is possible. In 2025, two independent studies inventoried more than

5,000 proteins per cell – a figure that is already being surpassed. Olsen, who led one study⁴, says his group has work under review in which they identify up to 7,000 proteins in individual cells. And Karl Mechtler, a proteomics researcher at the Research Institute of Molecular Pathology in Vienna, who co-directed the other study⁵, notes that with some larger cells, “you can get up to 8,000 or 9,000 proteins”.

Many SCP laboratories now rely on the cellenONE, a device commercialized by Cellenion in Lyon, France, that makes it possible to isolate individual cells from a culture or tissue sample while also imaging them. That allows users to record each cell's shape and size, which matter because cells can vary considerably in volume and protein content.

Downstream processing steps have also been optimized, minimizing sample loss and preventing cross-contamination that could confound interpretation of the data. Although chemical labelling of proteins was a key step in Slavov's 2018 publication, many groups have adopted unlabelled approaches that are simpler and faster. For example, Kelly's team developed a single-step process in 2023 that remains its go-to protocol⁶. And Chen and her collaborator at Academia Sinica, Hsiung-Lin Tu, have integrated the full sample preparation process into a compact microfluidic device. Known as the SciProChip, it performs all the necessary steps in nanoscale volumes of liquid⁷. “The chip substantially improved our sensitivity, reproducibility and quantitation accuracy,” says Chen.

Advances in MS instrumentation have also propelled the field. Popular options include the Orbitrap Astral machines from Thermo Fisher Scientific in Waltham, Massachusetts, and the timsTOF Ultra line from Bruker in Billerica, Massachusetts. These systems employ different strategies to detect and quantify proteins that would be invisible with older

technology, but can cost around US\$1 million. “We can quantify much more sensitively, much more deeply, and at high throughput even more proteins in single cells,” says Slavov. However, he notes that the more affordable, older-generation systems remain effective for profiling high-abundance proteins.

The biggest hurdle is speed. Most SCP pipelines top out at a few hundred cells per day – orders of magnitude lower than the throughput of transcriptomics studies. This is mainly because of the need to separate cellular peptides by size before MS analysis. Going faster can mean sacrificing precious data, says Kelly: “When we switch from 288 samples per day to 500 samples per day, we do see a big hit in our proteome coverage.”

Multiplexing offers one solution, in which individual cells are labelled such that the data from each can be disentangled even after the cells have been pooled for processing and analysis. For example, the labelling strategy Slavov used in 2018, based on MS-resolvable ‘tandem mass tags’, can resolve dozens of samples in parallel. But labelling adds complexity to sample preparation and can confound data analysis. “In the end, you end up getting actually quite a low identification rate from the peptides compared to a label-free experiment,” says Olsen.

In 2023, Slavov and his former graduate student Harrison Specht co-founded the Parallel Squared Technology Institute in Watertown, Massachusetts, to develop new multiplexing strategies to boost SCP throughput. In 2025, the institute described a two-pronged approach that combines MS-detectable mass labels with a ‘timePlex’ method that staggers the introduction of samples into the spectrometer⁸. “We’ve already demonstrated 27-plex and we can easily see this going to more than 100-plex,” says Slavov. Bearing that out will require testing and validation by the community. For now, Mechtler says, “multiplexing is a need, but we don’t have the answer”.

Mysteries resolved

Even with the existing speed limit, early adopters and pioneers have made considerable progress in demonstrating the practical utility of SCP. Slavov’s collaboration with Zernicka-Goetz and Chou helped to crack a mystery of developmental biology that would have been impossible to resolve without proteomic data, showing that newly formed embryos have already established patterns that do not manifest until later in development. In those earliest stages of development, notes Slavov, “the genome is not actively transcribed [and] many of the proteins come from the mother” – that is, they’re inherited, rather than made by the fertilized egg.

Van Eyk’s team has been using SCP to understand how variability in the cells that form

cardiovascular tissue contributes to heart disorders and how these respond to therapy. In a preprint posted in May, the researchers treated stem-cell-derived heart cells known as cardiomyocytes with two different drugs and identified cardiomyocyte subtypes with strikingly divergent responses and dose sensitivities⁹. “If you have a clinical trial and someone doesn’t respond, it could be that your drug is not hitting the right pathway in the target cell,” says Van Eyk, “or it could mean that it is hitting the right target cell, but the proteome of that target cell is in a different state.”

Single-cell details could also provide a direct mechanistic window into why some tumours are more lethal than others. In 2025, Mechtler was part of a team that used SCP to profile immune cells from people with glioblastoma, an aggressive form of brain cancer¹⁰. The team identified distinct cell subtypes that either promote or prevent tumour growth – functional distinctions that would have been invisible in a transcriptomic analysis, because they arise from existing proteins that become

“We were able to identify druggable targets from individual cells.”

active in the cell rather than newly activated gene expression. Such insights could inform therapies that tilt the immune system towards a potent antitumour response.

‘Spatial’ is the other big buzzword in contemporary biology, and Mann’s team is finding ways to perform deep proteomic profiling of individual cells while preserving knowledge of their location. His group’s ‘deep visual proteomics’ strategy relies on laser microdissection to precisely carve out individual cells or even subsections of cells for MS analysis, with artificial-intelligence algorithms to aid cell segmentation and selection. In 2024, he and his collaborators used this method to untangle the mechanism underlying a dermatological condition known as toxic epidermal necrolysis¹¹, which can be fatal. “We found which cells this happens in and what’s the culprit,” he says, “and then we did a drug repurposing so that the disease is now treatable, so nobody has to die of it any more.”

Closing in on completion

At the current rate of progress, near-complete single-cell proteomes could soon be within reach. Kelly estimates that coverage of around 12,000 distinct protein types might be enough to capture the diversity of proteins in a cell. But this will take some doing. “If you detect 1,000 proteins in your analysis in proteomics in general, it tends to be the most abundant thousand,” says Kelly. By extension, the missing pieces of the proteome will be the scarcest,

and it will take a substantial boost in sensitivity and analytical performance to fill in the blanks.

Such deep proteome coverage could be highly informative, however, encompassing low-abundance variants and proteins that have undergone chemical alterations such as phosphorylation, which can substantially alter function and stability. Protein molecules bearing such ‘post-translational modifications’ are much less abundant than are their unmodified counterparts – Chen estimates that less than 1% of the proteome has been phosphorylated, for example. And it remains difficult to discern these variants with current MS workflows.

Chen and Tu have modified their SciProChip to isolate phosphopeptides specifically and evaluate how their abundance changes in cancer¹². “We were able to identify druggable targets from individual cells, demonstrating the potential of single-cell phosphoproteomics to advance precision medicine,” says Chen. She is now exploring opportunities to use this technology to guide treatment selection.

That said, many practitioners are focusing more on building robust and reproducible SCP pipelines than pushing the limits of detection. The technology is costly and there are many opportunities for things to go wrong during an experiment. “You have different challenges like the sample preparation, the chromatography, the mass spec,” says Mechtler. “If something is wrong, you will completely fail, and sometimes you have no idea what’s going on.” As such, collaboration with an SCP expert group might be the easiest way for interested researchers to dip their toes into the water.

To help to build those bridges, Slavov’s lab has for the past nine years hosted an annual conference on SCP, and he is hopeful that more people will recognize the method’s capabilities. “It’s a highly quantitative – but less accessible – tool that should be used for challenging and meaningful things such as determining molecular mechanisms,” Slavov says. “Single-cell proteomics can do this better than almost any other method that I can think of.”

Michael Eisenstein is a science writer in Philadelphia, Pennsylvania.

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